

Commission & Consolidator Payouts

Module specification for CRM integration

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This module answers one question on every ticket: **how much will the host agency's cheque be, and is it right**. It reads the ticket, finds the contract that governs it, computes the figure, and pre-fills the commission field. Once a week the host's statement is uploaded and the module reconciles what was paid against what was owed.

It is built as a **pure function**: text in, JSON out. No database calls, no clock, no network. The same ticket always produces the same number, and the same code runs inside the CRM, in a queue worker, in an Edge Function, and in tests.

Everything here has been checked against real documents - two signed contracts, a real weekly statement, and real AIR files. Where a rule is inferred rather than stated, it says so. Where a real document contradicted an earlier assumption, the contradiction is recorded rather than quietly corrected, because the same mistake is easy to make twice.

Scope for the first build

EL AL (LY) under Main St. Travel only. That is where the volume is, and one airline under one host proves the whole path end to end. The data model already keys on tenant, host and airline, so a second host is new rows rather than a rewrite - but do not build ten before one works.

1 - The three figures

Every ticket produces three numbers, in this order. They must not be collapsed into one.

#	Figure	What it is
1	Commission	What the airline pays, less the host's share. Zero on a net or bulk fare, and zero on a published fare that fails a contract condition.
2	Fees	What the host charges: per exchange, per refund, on a net fare, on a ticket that earned nothing. Always negative.
3	Markup	On a net fare, the agent's own margin - selling fare less net fare. It is the agent's commission on that ticket, self-set inside a ceiling the airline permits.

The cheque

$$\text{cheque} = \text{commission} + \text{markup} - \text{fees}$$

The markup belongs in the cheque. The whole ticket, markup included, is collected under the host's accreditation and settles to them - so the markup is not money the agent already holds, it is money the host is holding on the agent's behalf and has to remit. From the agent's side there is one figure, and this is it.

The single most expensive thing to get wrong. Reconciling a statement against the COMMISSION alone reports every marked-up bulk ticket as paid-where-nothing-was-due, and a bulk ticket missing from the statement as agreement rather than a shortfall. On the sample batch that is the difference between an expected total of \$89.76 and \$3,245.26 - the gap is markup being silently written off. Reconcile against the cheque.

Worked, on real tickets

Ticket	Fare	Base	Commission	Fee	Markup	Cheque
bulk business	net	12,378.75	0.00	-123.79	2,475.75	2,351.96
marked-up economy	net	3,398.75	0.00	-33.99	679.75	645.76
published S	published	1,496.00	0.00	-10.00	0.00	-10.00
exchange	net	3,707.00	0.00	-25.00	0.00	-25.00

A ticket can be **negative**: it earned no commission and still cost a fee. That is a real outcome, not an error, and the UI has to show it.

2 - Net fares: the rule that governs them

On a net fare the airline files a fare it will accept and lets the agent sell above it. The agent's margin is their commission on that ticket. The host's fee follows one principle:

The host cannot be worse off because the agent chose to issue net. Whatever the host would have earned had the same fare gone out published with commission, they earn. The agent's choice of how to take their margin is theirs, and it is not allowed to cost the host anything.

So the host's fee on a net fare is **the higher of** the point they keep on a published fare and a flat figure by cabin. The cabin figure is a FLOOR, not the fee.

net fare fee = max(published-fare point x base fare, cabin floor)

12,378.75 business: 1% = 123.79 floor 50 -> 123.79
 900.00 economy: 1% = 9.00 floor 15 -> 15.00

Reading the cabin figure as the fee rather than as a floor understates the host on every large fare - \$50 where \$123.79 is owed on the sample business ticket - and would have the reconciler reporting the host's own correct deduction as an unexplained one.

The markup ceiling

The airline permits a markup up to a limit, which varies by fare. Where that limit is known and configured, the module reports the markup as a percentage and the **headroom** left under it. Headroom is the one loss no statement can show: marking up 10% where 25% was permitted leaves money behind, nobody short-paid anything, and reconciliation says AGREES. An unconfigured ceiling reports as **unknown**, never as zero.

Reading the markup from the file

Amadeus records it in the FM element as FM*G*679.75A. Those digits match the commission-amount pattern exactly, so **the *G* must be tested first** - otherwise the agent's own margin is filed as commission the host owes, and every marked-up ticket reports a shortfall. Cross-check against selling base less net base (KS- less KN-): the two are independent and must agree. On both real records they agree exactly.

3 - Finding the contract

Every ticket carries the IATA number it was issued under. **The IATA number owns the contract, not the agency** - one host commonly holds several numbers with different rates on the same airline. The MST agreement says so itself: "PNRs created under one of MST's affiliate offices in order to access higher contracted commission levels."

#	Step	If it fails	What that means
1	Tenant	reject	Never price across tenants
2	IATA on the ticket	no_iata	Nothing to match on
3	Office for that IATA	unknown_iata	A contract to go and get
4	Contract for the airline	no_carrier	A contract to negotiate
5	Ticketing date in window	outside_window	A renewal
6	Clause: route, class, fare type, tour code	NO_RULE	Contract held, nothing covers this

Steps 1-5 are the **fallback**: no contract could be chosen, so no figure is written. Step 6 is different - a contract exists and no clause covers the ticket.

NIL and NO_RULE must never merge. NIL means a clause asserted zero - EL AL forfeits commission when the mandatory tour code is missing, and that is a settled answer. NO_RULE means nobody has worked it out. Both display as \$0.00; only NIL may be written into a field. Merging them books zero on tickets that were actually owed money.

The folder is not all yours

In a 15-file sample drawn from the live airfile folder, **9 belonged to a different agency** - IATA 33770903, a separate travel business, with its own passengers and its own contracts. Any total computed across the folder without filtering by IATA is wrong, and wrong in a direction that looks plausible. **Filter by IATA before anything else.**

Bootstrapping from the files

discoverFromFiles() groups incoming documents by IATA and carrier and reports ticket counts, date ranges, fare types, tour codes, and the rates actually claimed per booking class - flagging each group as configured or not. Use it to populate the contract table from history, and to see at a glance which IATA numbers are in the folder.

It proposes a rate only where **every** ticket in a class agreed on one. Where they disagree it reports the disagreement instead: what an agent claimed is evidence, not a contract, and seeding a table with the more popular of two wrong answers would encode the mistake as policy and reconcile against it forever.

4 - Data model

Everything is editable, contracts are uploadable, and every query is scoped by tenant.

Table	Key columns
tenant	id, name. Every query below is scoped by tenant_id.
office	id, tenant_id, iata (the join key, unique per tenant), name, agency (label grouping several offices under one host), retains_points, fee_schedule
carrier_contract	id, office_id, carrier , issued_from, issued_to, rates (JSON: class -> percent), include_yq, required_tour_code, origin_in, scope, max_markup_percent, markup_basis, files[]
host_agreement	id, office_id, the fee schedule: per-fare-type splits and the fee rows (exchange, refund, net fare by cabin, non-commissionable). Each row carries approved: boolean .
ticket_document	One row per passenger per document: ticket number, type, issue date, carrier, iata, base fare, net fare, taxes, coupons, fare type, tour code, plus the computed commission, fees, markup, cheque and outcome.
statement_line	One row per line of an uploaded statement: invoice number , ticket number (usually null), passenger, vendor, invoice date, travel date, itinerary, signed amount, remark.

Keep the approved flag. Clauses that depend on facts an AIR file does not carry - corporate bookings, promotional net fares, discretionary minimum fees - are encoded but left unapproved, so the engine surfaces them as exposure rather than booking them as certainty. An unapproved clause never spends.

5 - The pre-fill rule

The module returns prefill: a decimal string, or null. **Bind it straight to the field.** The decision about whether a number is safe to write has already been made.

Outcome	prefill	Meaning
CALCULATED	the amount	A clause matched and produced a figure
NIL	0.00	A clause asserted nothing is due - a settled answer
NO_RULE	null	Contract held, no clause covers this ticket
AMBIGUOUS	null	Two clauses matched equally. Never guessed
INCOMPLETE	null	A clause needed something the document lacked
NO_CONTRACT	null	No contract for this IATA - the fallback
ERROR	null	Could not be priced at all

Two further conditions suppress a pre-fill even on a good outcome:

Condition	Why
The layer being paid must be the layer that settled	On a Zurich-Tel Aviv reissue the carrier layer resolved cleanly (a \$100 clawback) while the sub-agent layer could not resolve its share. Reading one and paying the other offered a confident \$0.00 on an open question.

Condition	Why
No REVIEW flag on the document	REVIEW is raised when two signed contracts disagree - on that same ticket, that the host pays \$100 out of pocket. A figure already flagged for a human is not one to bind to a field.

Every result carries explanation, clause and ruleid. Show the explanation beside the field. A number the agent cannot trace to a line of the contract is one they cannot dispute, which defeats the point.

6 - The weekly statement

This section was rewritten after reading a real one. Every assumption the first version made about the format was wrong.

It is not a commission statement. It is an accounts ledger. One line per thing the host billed or credited, and the balance at the foot is the payout. On the statement read here: 149 lines, 141 invoices, charges \$1,643.00, credits -\$32,162.65, **payout -\$30,519.65.**

Three facts about the format

#	Fact	Consequence
1	The key is the INVOICE number, not the ticket number. 11 of 149 lines carry a ticket number and every one of those 11 is \$0.00 - they are the airline's own issue records. Every line carrying money has no ticket number at all.	A reconciler matching on ticket number matches nothing . See §7.
2	Sign carries the meaning. Negative is a credit the host owes the agent - markup and commission. Positive is the host's fee.	Sum the signed amounts; a negative total is what the host owes.
3	One invoice can carry several lines , and the host nets them itself, printing a subtotal on a line of its own. A \$50 credit against a \$25 exchange fee prints as -\$25.00.	Subtotal lines are restatements, not transactions. Counting them double-counts.

Parsing it

A fixed-width report, two lines per invoice, six pages - each page carrying its own header at **different column offsets**, and rows that do not reliably align to their own page's header. So fields are recovered by what they are, not where they sit: a ticket number is a run of digits at the front, a date is a date, the passenger carries a "/", the vendor is what sits between them.

Validate against the statement's own bottom line. The reader sums its lines to exactly the balance the statement prints at its foot, -\$30,519.65, with no warnings. That is a check against arithmetic nobody on our side did, and it is what makes every other figure believable. Keep this assertion in the test suite.

Two bugs it caught, both of which produced plausible wrong numbers rather than errors: "Total Open" and "Account Balance" each restate the balance, and counting them as transactions **tripled** the payout to -\$91,558.95; and 8/16/2026 was being read as the 8th of the 16th month.

7 - Matching tickets to statement lines

This is the hardest part of the module and the part most likely to be got wrong quietly. The statement's money lines carry **no ticket number**, so the join has to be made on what is there: passenger, itinerary and travel date.

The matching rule

```
match = surname equal
      AND travel date equal
      AND given name compatible (statement truncates it)

no confident match -> report as unmatched, never guess
```

Surname alone is not enough, and the failure is silent. A first pass matching on surname produced eleven "discrepancies" on eight tickets: Klein/Joel paired with Klein/Esther, Landau/David with a different Landau trip, Raab against a booking from a month earlier. All false. Adding the travel date cut it to two confident matches and three honest gaps. **Two real findings beat eleven that fall apart when checked.**

What the first real reconciliation found

Eight of the agent's tickets fell inside the statement period. Two matched a statement line confidently. **Both disagreed with the contract, and in the same direction.**

Passenger	Ticket	Cls	Fare	Contract says	MST charged	Gap
AHARONI/SHOSHAN	...4473	I	6,085.00	-10.00	-50.00	40.00
AHARONI/YEHUDA	...4478	Q	2,714.00	-10.00	-30.00	20.00

Both are **published** fares (K-F, no SIAB;;BT) that earned no commission for want of the tour code. The schedule prices that at a flat **\$10**. MST charged **\$50** and **\$30** - which are exactly the schedule's *net-fare cabin* figures (\$30 premium, \$50 business).

Open question, and the most valuable one in this document. Either MST applies the cabin fees to published fares too, or the net-fare row is being read wrongly, or these are net fares the AIR file does not mark. It is \$60 on two tickets - but it is a rule that applies to **every zero-commission ticket**, and none of the sampled tickets carry the tour code. Put this to MST before phase 2.

Six of the agent's tickets had no statement line at all - three children on one family booking (the fourth passenger was billed), and three passengers on another. On the current reading those cost \$10 each, so not being billed is \$60 *in the agent's favour*. The fee is not being applied consistently in either direction, which is itself the finding.

8 - Reconciliation output

Each line lands in exactly one of eight states. Nothing is dropped: a statement line matching no ticket, and a ticket on no line, are both findings.

Status	Severity	Meaning	Action
SHORT_PAID	critical	Paid less than the contract gives	Claim the difference

Status	Severity	Meaning	Action
NOT_ON_STATEMENT	critical	Owed, and the statement omits it	Claim in full
OVER_PAID	warning	Paid more than the contract gives	Expect a clawback
PAID_WHERE_NONE_DUE	warning	Paid on a document owed nothing	Verify before spending
NOT_IN_TICKETS	warning	Paid for a document not in our records	Investigate
DEDUCTION	warning	A fee withheld no agreement covers	Query the fee
UNMATCHED	warning	No confident join to a ticket	Match by hand
AGREES / CORRECTLY_NIL	ok	Both sides agree	-

Excel export

One row per ticket, sorted **worst first**. Variance is the money column; negative means underpaid. Red fill on negative, amber on positive, freeze the header.

Column	Example	Column	Example
Status	OVER_CHARGED	Commission	0.00
Ticket	114-7504084473	Fees	-10.00
Passenger	AHARONI/SHOSHAN	Markup	0.00
Travel date	2026-08-05	Cheque due	-10.00
Route	MIA-TLV	MST invoice	10109752
Class / fare	I / published	MST amount	-50.00
Base fare	6085.00	Variance	-40.00
Match basis	surname+date	Why	nil: tour code required

Add a totals row splitting variance into *short-paid*, *over-charged*, *missing* and *unexplained deductions*, plus a count of unmatched rows on both sides. Those five figures are the week.

9 - API

```
import { priceAirFile, parseMstClientStatement, discoverFromFiles }
  from '@commission/sdk'

const result = priceAirFile(airText, {
  config, tenantId: 'acme', view: 'subagent',
})
```

```

for (const doc of result.documents) {
  doc.prefill          // '645.76' or null -> leave the field empty
  doc.outcome          // 'CALCULATED' | 'NIL' | 'NO_RULE' | ...
  doc.subAgentCommission // '0.00'      airline's commission, less host share
  doc.fees             // [{ clause, label, amount }]
  doc.markup           // '679.75'      the agent's own margin
  doc.totalToSubAgent   // '645.76'      THE CHEQUE - reconcile against this
  doc.markupPercent     // '25.0000' or null if no ceiling configured
  doc.markupHeadroom    // '0.00' = at the ceiling; negative = over it
  doc.explanation, doc.clause, doc.flags
}

const st = parseMstClientStatement(statementText)
  st.lines[]           // invoice, ticketNumber (usually null), passenger,
                        // vendor, issueDate, startDate, itinerary, amount
  st.invoices[]         // grouped, with net and the printed subtotal
  st.totals.payout      // our sum
  st.totals.statedBalance // the statement's own - ASSERT THESE MATCH

```

All money crosses the boundary as decimal strings - no bigint, no floats, no Date - so the whole response survives JSON.stringify. An HTTP wrapper is provided (POST /price, POST /statement) with no dependencies.

10 - Known gaps

Found on real files. Fix before trusting a full-folder run.

Gap	What happens	Fix
EMDs unrecognised	A baggage EMD reads as a TKT with ticket number UNKNOWN, carrier ??, fare 0.00. Seen twice in a 15-file sample.	Detect EMD/TMCD and type it as EMD
Validating carrier missed	Where the A- line is empty the carrier reads ??, so no contract resolves. Seen on United and Arkia tickets.	Fall back to the FV element
Foreign currency fare	K-FPLN6789.00;USD1811.00 is read as PLN 6,789. The statement settles in USD, so the ticket prices in the wrong currency or throws a mismatch.	Take the second field, the settlement currency
Other agencies in the folder	9 of 15 sampled files belonged to IATA 33770903. Totals across the folder are wrong.	Filter by IATA first, and surface the others as a finding

11 - Build order

Phase	Scope	Done when
0	Fix the four gaps in §10. They are all on real files.	A full-folder run produces no UNKNOWN tickets or ?? carriers

Phase	Scope	Done when
1	Contract tables + resolution. Read-only: compute and display.	Every LY/MST ticket shows a cheque figure and an explanation
2	Pre-fill. Write the number into the field.	Agents stop entering commission by hand on CALCULATED and NIL tickets
3	Statement upload, matching, reconciliation, Excel export.	A week's statement produces a sorted exception list
4	Contract editing and PDF upload in the UI.	A new airline can be added without a deploy
5	Second host agency; then standalone for other agencies.	The engine has no CRM dependency - packaging, not a rewrite

Phase 1 being read-only is the important one. It lets the computed figure be compared against what agents actually entered, across thousands of historical files, before a single number is written automatically. That comparison is the cheapest validation available - it is how a ticket claiming 8% on a class EL AL files at 7% was found.

Non-negotiables

Rule	Reason
Money is integer minor units, never floats	A half-cent bias is invisible per ticket and material across a year
NIL and NO_RULE never merge	Both show \$0.00; only one is an answer
Reconcile against the cheque, not the commission	Otherwise every marked-up ticket reports wrongly
Match on surname AND travel date, or not at all	Surname alone produced eleven false discrepancies out of eleven
Filter by IATA before any total	The folder holds other agencies' files
Assert our statement sum equals the statement's own balance	It is the only external check on the reader
A tie is never broken arbitrarily	Two clauses matching equally returns AMBIGUOUS
Every figure cites its clause	The output is evidence in a dispute
Unapproved clauses do not spend	A term nobody confirmed is exposure, not income

Open questions for Main St. Travel

Question	Impact
Why was a published, non-commissionable fare charged \$50 and \$30 rather than the schedule's \$10? Both are the net-fare cabin figures.	Applies to every zero-commission ticket; \$60 on two sampled
Is the commissionable basis the base fare alone, or base plus YQ/YR? Neither contract states it.	~\$35 per ticket on a long-haul fare
Is the host's point on a NET fare struck on the selling fare or the net fare before markup?	\$33.99 vs \$27.19 on one sampled ticket
Which booking classes count as Economy, Premium and Business for the cabin floor?	Decides a \$15 floor from a \$50 one
Is the tour code enforced in practice? No sampled ticket carries it.	Forfeits the commission on every published fare
Why do some tickets on a family booking get no statement line at all?	Six sampled tickets unbilled; direction varies
Statements are weekly, settlement is monthly - which is authoritative?	Whether a weekly shortfall is real or timing